## Question 4 - Part 2

$$J(b_0, b_1) = \frac{1}{n} \sum_{i=1}^{n} (\tilde{y}_i - (b_0 + b_1 \tilde{x}_i))^2 + \alpha b_1^2$$

$$\frac{\partial J}{\partial b_0} = \frac{-2}{n} \sum \tilde{y}_i - b_0 - b_1 x_i = 0$$

$$\frac{\partial J}{\partial b_0} = \frac{-2}{n} \left( \sum \tilde{y}_i - \sum b_0 - \sum b_1 x_1 \right) = 0$$

given $\sum \tilde{y}_i$ and $\sum \tilde{x}_i$ equal 0

$$\therefore \sum b_0 = 0 \rightarrow n b_0 = 0 \rightarrow b_0 = 0$$

Plug in $b_0 = 0$

$$\frac{\partial J}{\partial b_1} = \frac{-2}{n} \sum \tilde{x}_i (\tilde{y}_i - b_0 - b_1 \tilde{x}_i) + 2\alpha b_1 = 0$$

$$-\sum \tilde{x}_i \tilde{y}_i + b_1 \sum \tilde{x}_i^2 + n\alpha b_1 = 0$$

$$b_1 \left( \sum \tilde{x}_i^2 + n\alpha \right) = \sum \tilde{x}_i \tilde{y}_i$$

$$\boxed{b_1 = \frac{\sum \tilde{x}_i \tilde{y}_i}{\sum \tilde{x}_i^2 + n\alpha}}$$

# Question 4 - Part 4

$$RSS = \frac{1}{n}\sum_{i=1}^{n}(\tilde{y}_i - b_1\tilde{x}_i)^2$$ * given $b_0 = 0$ from pt. 2

$$\frac{\partial RSS}{\partial b_1} = -\frac{2}{n}\sum \tilde{x}_i(\tilde{y}_i - b_1\tilde{x}_i)$$

when $b_1 = 0$:

$$\frac{\partial RSS}{\partial b_1} = -\frac{2}{n}\sum \tilde{x}_1\tilde{y}_1$$

The subgradient of the slope is any value from $-1$ to $1$, so the penalty must have an equal or greater magnitude to reduce the entire gradient to 0. Therefore $b_1 = 0$ is only optimal if:

$$\boxed{\alpha \geq \left|\frac{2}{n}\sum_{i=1}^{n}\tilde{x}_1\tilde{y}_1\right|}$$